MIT OpenCourseWare

7.013

Introductory Biology 7.013 — Introductory Biology May 2026 **EXTREMELY**

HARD – MIT LEVEL

Time Limit: 3 hours

Total Marks: 100

1. (2 points) Construct an original proof-level problem involving Genomics for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

2. (2 points) Construct an original proof-level problem involving Gene Regulation for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

3. (5 points) Prove the fundamental theorem concerning Stem Cells in the context of 7.013 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

4. (3 points) Analyze the limitations of standard approaches to Cancer Biology when applied to edge cases in 7.013. Construct explicit counterexamples showing where intuitive reasoning fails.

5. (2 points) Prove the fundamental theorem concerning Neurobiology in the context of 7.013 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

6. (6 points) Analyze the limitations of standard approaches to Genomics when applied to edge cases in 7.013. Construct explicit counterexamples showing where intuitive reasoning fails.

7. (5 points) Analyze the limitations of standard approaches to Gene Regulation when applied to edge cases in 7.013. Construct explicit counterexamples showing where intuitive reasoning fails.

8. (4 points) Prove the fundamental theorem concerning Stem Cells in the context of 7.013 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

9. (10 points) Construct an original proof-level problem involving Cancer Biology for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

10. (3 points) Construct an original proof-level problem involving Neurobiology for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

11. (3 points) Construct an original proof-level problem involving Genomics for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

12. (4 points) Prove the fundamental theorem concerning Gene Regulation in the context of 7.013 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

13. (3 points) Construct an original proof-level problem involving Stem Cells for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

14. (5 points) Analyze the limitations of standard approaches to Cancer Biology when applied to edge cases in 7.013. Construct explicit counterexamples showing where intuitive reasoning fails.

15. (2 points) Analyze the limitations of standard approaches to Neurobiology when applied to edge cases in 7.013. Construct explicit counterexamples showing where intuitive reasoning fails.

16. (3 points) Construct an original proof-level problem involving Genomics for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

17. (5 points) Prove the fundamental theorem concerning Gene Regulation in the context of 7.013 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

18. (4 points) Prove the fundamental theorem concerning Stem Cells in the context of 7.013 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

19. (3 points) Construct an original proof-level problem involving Cancer Biology for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

20. (4 points) Prove the fundamental theorem concerning Neurobiology in the context of 7.013 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

21. (2 points) Analyze the limitations of standard approaches to Genomics when applied to edge cases in 7.013. Construct explicit counterexamples showing where intuitive reasoning fails.

22. (7 points) Analyze the limitations of standard approaches to Gene Regulation when applied to edge cases in 7.013. Construct explicit counterexamples showing where intuitive reasoning fails.

23. (3 points) Construct an original proof-level problem involving Stem Cells for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

24. (4 points) Construct an original proof-level problem involving Cancer Biology for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

25. (6 points) Construct an original proof-level problem involving Neurobiology for 7.013 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

Solutions

Solution 1:

Solution approach: First recall the definitions and key theorems related to Genomics from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 2:

Solution approach: First recall the definitions and key theorems related to Gene Regulation from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 3:

The proof follows from the definitions and properties established in 7.013. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 4:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cancer Biology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 5:

The proof follows from the definitions and properties established in 7.013. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Genomics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Gene Regulation. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 8:

The proof follows from the definitions and properties established in 7.013. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 9:

Solution approach: First recall the definitions and key theorems related to Cancer Biology from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 10:
Solution approach: First recall the definitions and key theorems related to Neurobiology from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 11:
Solution approach: First recall the definitions and key theorems related to Genomics from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 12:
The proof follows from the definitions and properties established in 7.013. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 13:
Solution approach: First recall the definitions and key theorems related to Stem Cells from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 14:
The edge case analysis reveals the subtle assumptions underlying standard treatments of Cancer Biology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 15:
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The edge case analysis reveals the subtle assumptions underlying standard treatments of Gene Regulation. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 23:

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Solution 24:

Solution approach: First recall the definitions and key theorems related to Cancer Biology from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 25:

Solution approach: First recall the definitions and key theorems related to Neurobiology from 7.013. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.